

Taylor Series with Verbatim-app

Purpose

We've created a one-stop application that collects a student's conversation with a prompt that the instructor provides and generates a transcript formatted specifically for further analysis. The app is a single html file that can be loaded locally or hosted on a school server. A companion app, verbatim-setup, enables teachers to modify activity directions and items from an optional exit survey.

Features

The app includes a number of useful features including: (a) the ability to use various genAI, including local, free, and paid options; (b) a built-in profiler that determines student learning preferences; (c) built in math symbol template; (d) built-in after-activity survey; and (e) report exporting tools. Each of these are discussed in more detail below.

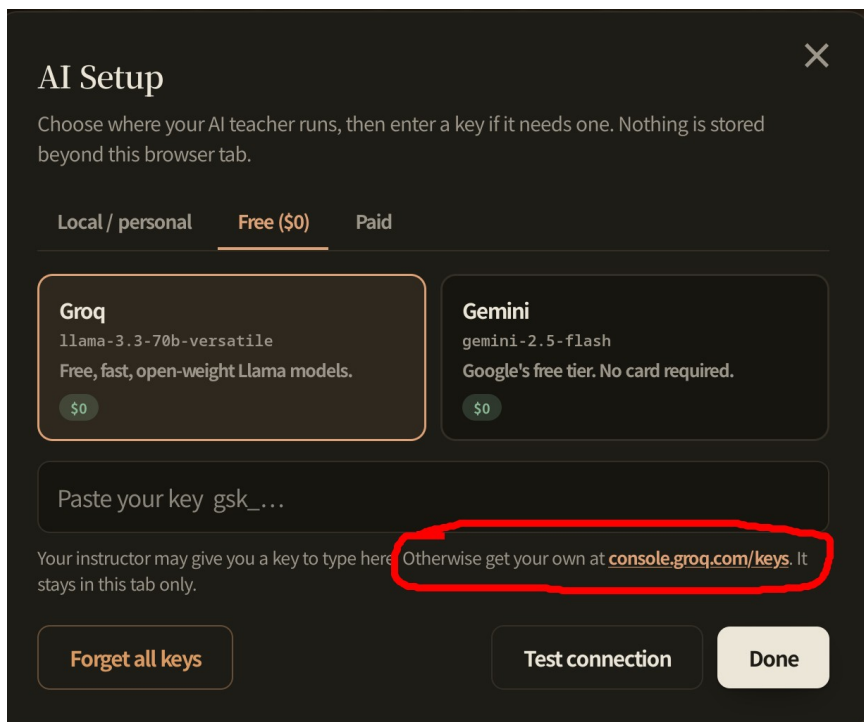
Various API KEY options. The application connects to the student's genAI of choice using an API Key. All discussions are funneled through the app for consistent formatting across models.

Students can choose models at different price levels. The app quietly checks which AI services answer on the room's network and hides the ones that don't. Verbatim-app records which services answered, never anything about the student or their location.

The screenshot shows the 'AI Setup' window with a close button (X) in the top right. Below the title, it says 'Choose where your AI teacher runs, then enter a key if it needs one. Nothing is stored beyond this browser tab.' There are three tabs: 'Local / personal', 'Free (\$0)', and 'Paid' (which is selected and underlined). The 'Paid' tab displays six AI model cards in a 3x2 grid. Each card includes the model name, its identifier, a description, and a 'PAID' button. The models are: DeepSeek (deepseek-chat, '深度求索 • the model the original study used.'), Kimi (moonshot-v1-8k, '月之暗面 Moonshot AI.'), GLM (glm-4-plus, '智谱清言 Zhipu AI.'), Qwen (qwen-plus, '通义千问 Alibaba Cloud.'), OpenAI (gpt-5, 'ChatGPT's models. Bring your own key.'), and OpenRouter (deepseek/deepseek-chat, 'One key, many models.'). At the bottom of the window are three buttons: 'Forget all keys', 'Test connection', and 'Done'.

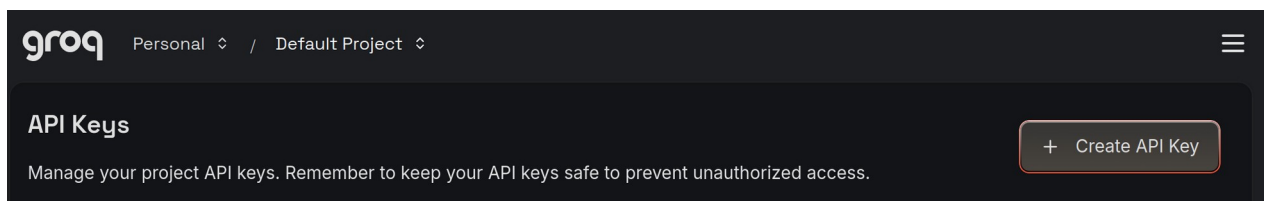
Model	Identifier	Description	Price Level
DeepSeek	deepseek-chat	深度求索 • the model the original study used.	Paid
Kimi	moonshot-v1-8k	月之暗面 Moonshot AI.	Paid
GLM	glm-4-plus	智谱清言 Zhipu AI.	Paid
Qwen	qwen-plus	通义千问 Alibaba Cloud.	Paid
OpenAI	gpt-5	ChatGPT's models. Bring your own key.	Paid
OpenRouter	deepseek/deepseek-chat	One key, many models.	Paid

For instance, to connect to a free (\$0) option, click on the Free (\$0) tab. I clicked that tab and then selected Groq from the options (note: Options depend upon what network options you have available. For instance, Groq may not be available in China).



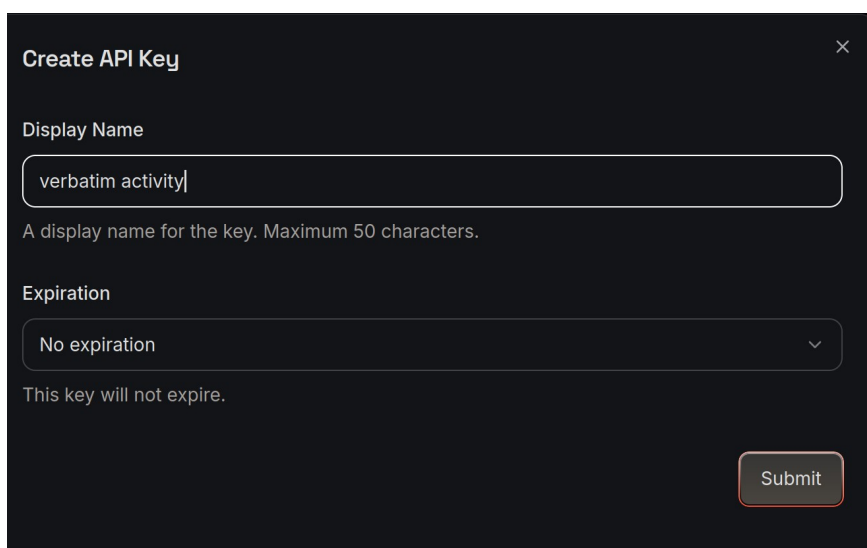
The 'AI Setup' dialog box has a dark background and a close button (X) in the top right. It contains three tabs: 'Local / personal', 'Free (\$0)' (which is selected), and 'Paid'. Under the 'Free (\$0)' tab, there are two cards. The 'Groq' card on the left lists 'llama-3.3-70b-versatile' and 'Free, fast, open-weight Llama models.' with a '\$0' badge. The 'Gemini' card on the right lists 'gemini-2.5-flash' and 'Google's free tier. No card required.' with a '\$0' badge. Below these cards is a text input field with the placeholder 'Paste your key gsk_...'. Underneath the input field is a line of text: 'Your instructor may give you a key to type here. Otherwise get your own at console.groq.com/keys. It stays in this tab only.' The link is circled in red. At the bottom, there are three buttons: 'Forget all keys' (orange), 'Test connection' (dark grey), and 'Done' (light grey).

Note that the dialog provides a link for obtaining an API-KEY for each option. For instance, here's a screenshot of the recommended Groq page (console.groq.com/keys).



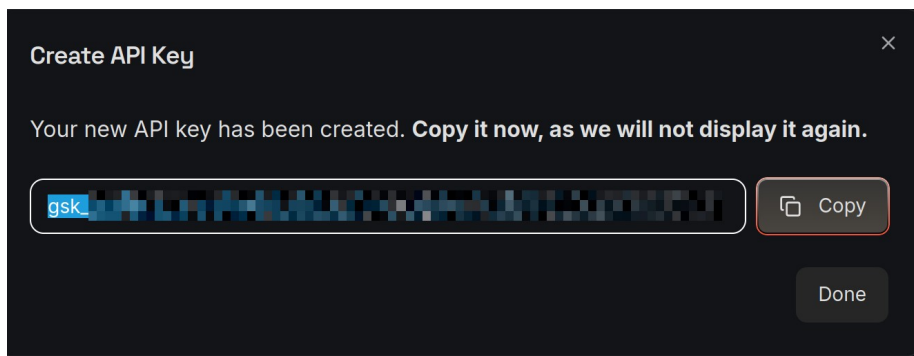
The 'groq' API Keys page has a dark header with the 'groq' logo, 'Personal' and 'Default Project' dropdowns, and a menu icon. The main section is titled 'API Keys' and includes the instruction 'Manage your project API keys. Remember to keep your API keys safe to prevent unauthorized access.' A '+ Create API Key' button is in the top right corner.

Click on the "Create API Key" button, type in a Display Name and set an expiration date. Then click SUBMIT to get a new key,

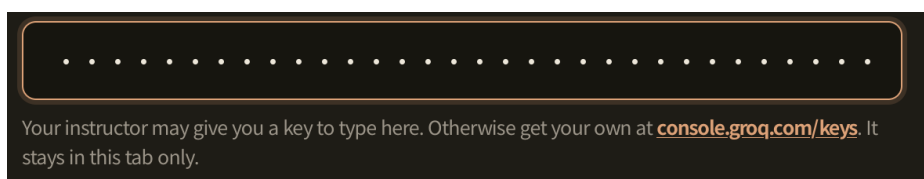


The 'Create API Key' form has a dark background and a close button (X) in the top right. It contains two sections. The 'Display Name' section has a text input field with 'verbatim activity' and a note: 'A display name for the key. Maximum 50 characters.' The 'Expiration' section has a dropdown menu set to 'No expiration' and a note: 'This key will not expire.' A 'Submit' button is at the bottom right.

A window with your API KEY will appear. You'll only see the code once, so copy it carefully then paste into the space provided in Verbatim-app.

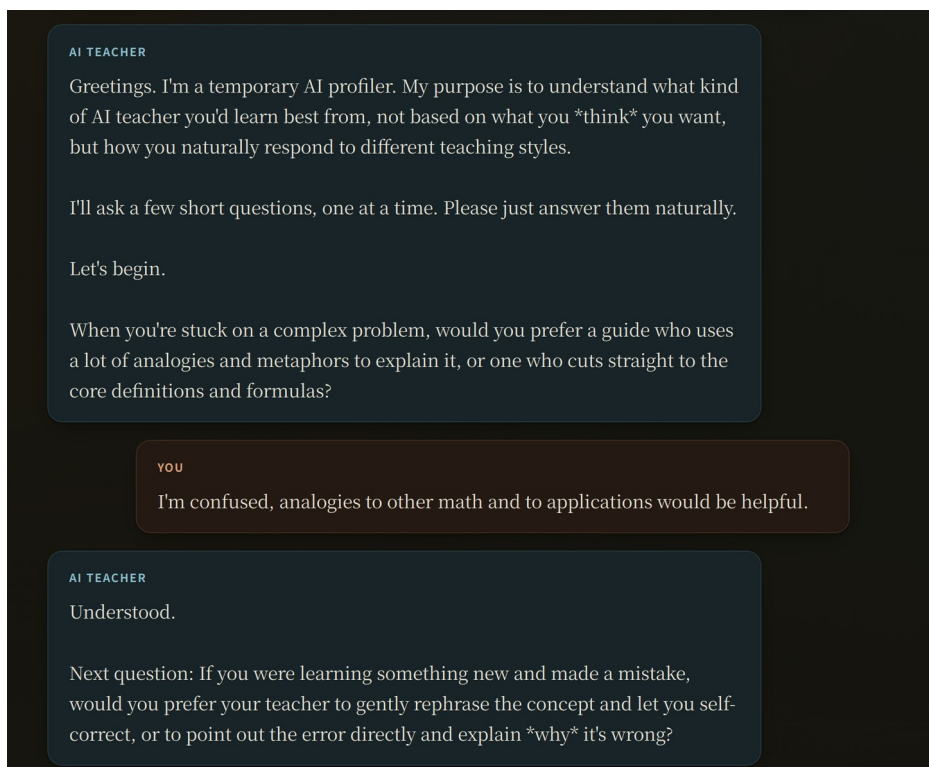


Your key is a long string of letters and numbers. When you paste into Verbatim, the key is replaced with dots and saved in your local computer until you delete it by clicking "Forget all keys" (see previous page). The key is like a password. Don't share it with others. Don't paste it into chat.

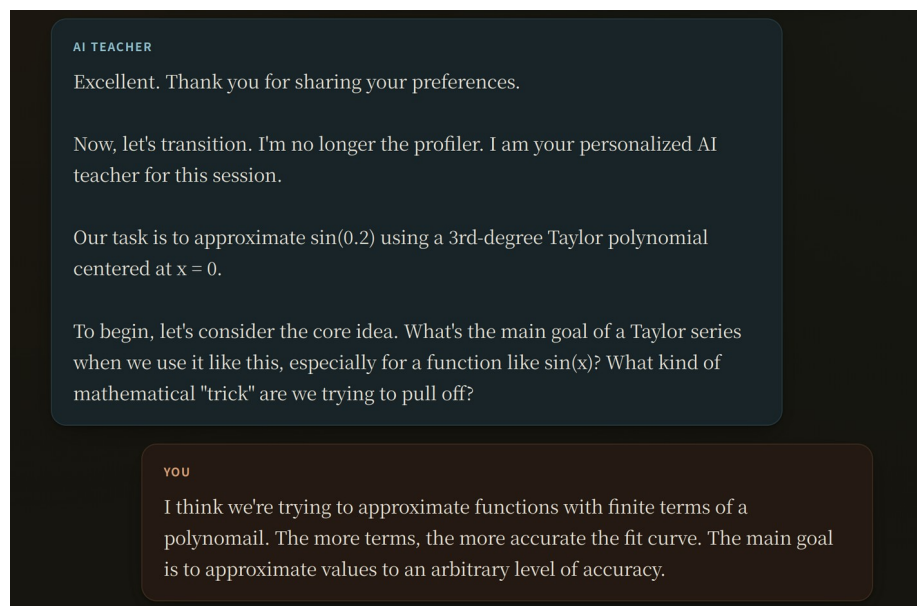


Note: The app also provides support for loading personal genAI models (from the Local/Personal tab), but that may take time and should be explored well in advance of data collection (e.g., a previous class meeting).

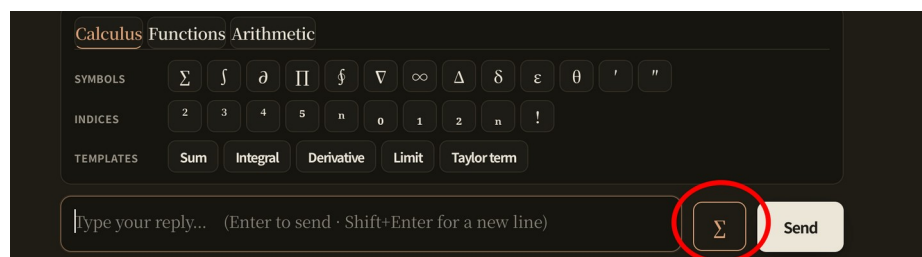
Profiler. Verbatim-app also supports the use of learning profiles. Our default instructions ask the AI to ask the student learning-profile questions. Student preferences are recorded in the chat and influence the AI's behavior during the chat.



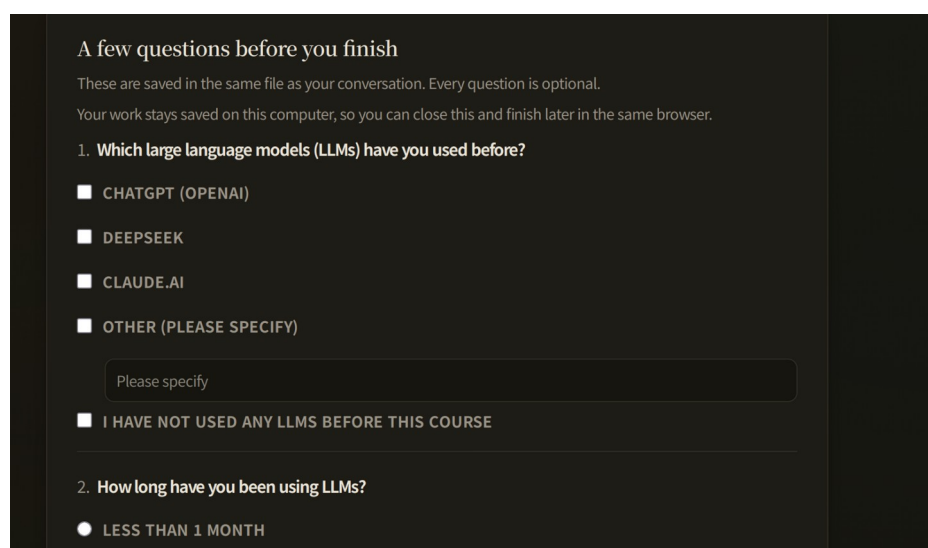
After the profile questions are completed, our default instructions direct the AI to facilitate a conversation about Taylor Series. (Note: Instructors can load different instructions using Verbatim-setup.)



The input bar comes with a palette of commonly-used mathematics symbols.



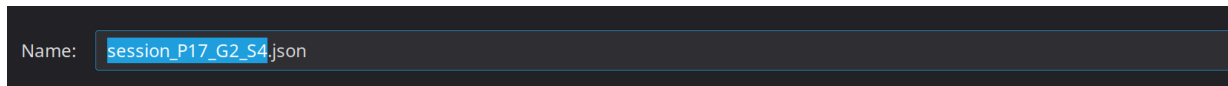
Built-In Survey. Once students complete the activity, they answer a survey about their experience within the app. Students are asked whatever questions are loaded into the app. The current questions match those that we used in the initial implementation of this study.



Questions may be changed by the instructor using Verbatim-setup.

Reporting. Once students complete the survey, they are provided with an option to download a report of the session. Reports are formatted specifically to ease data analysis. The reports include initial profiler

information, a complete transcript of the student/AI conversation (with student and AI clearly tagged), and the final survey questions. The default filename of the report includes student number and section to help keep data organized.



Here is a screenshot of the window students see when they download their work.

